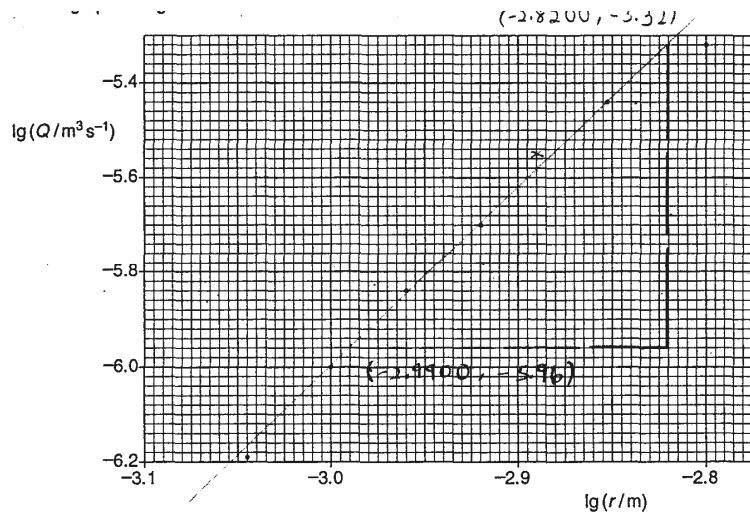


energy released may manifest as gamma radiation.

- 6 (a) When r is between $0.9 \times 10^{-3} \text{ m}$ and $1.5 \times 10^{-3} \text{ m}$, Q increases at an increasing rate with respect to r . When $r > 1.5 \times 10^{-3} \text{ m}$, Q increases with a decreasing rate with respect to r .

(b) (i)



Note: The last point is an anomaly. As such, the best fit line should be drawn based on the rest of the plotted points and there should be good scattering on both sides of the line.

(ii)
$$\text{gradient} = \frac{-5.32 - (-5.96)}{-2.8200 - (-2.9900)} \approx 3.76$$

(iii) Since $Q = kr^n$, we have $\lg Q = n \lg r + \lg k$. Substituting the gradient and one of the gradient coordinates, we have

$$-5.96 = (3.76 \times -2.9900) + \lg k$$

$$k \approx 1.98 \times 10^5$$

(c) (i) $k = \frac{\pi \Delta P}{8\eta L}$

$$1.98 \times 10^5 = \frac{\pi \times 500}{8\eta \times 0.20}$$

$$\eta \approx 4.96 \times 10^{-3}$$

(ii) Since $n = \text{gradient}$ and n is an integer, from (b)(ii), $n = 4$.

Using $Q = kr^n$, units of $k = \text{m}^3 \text{s}^{-1} \text{m}^{-4} = \text{m}^{-1} \text{s}^{-1}$.

Since $\eta = \frac{\pi \Delta P}{8kL}$,

$$\text{units of } \eta = \frac{\text{kgm}^{-1}\text{s}^{-2}}{\text{m}^{-1}\text{s}^{-1}\text{m}} = \text{kgm}^{-1}\text{s}^{-1}$$

(d) From (c), we have $\Delta P \propto k$. From (b), we have $k \propto \frac{1}{r^n}$.

Hence, $\Delta P \propto \frac{1}{r^n}$.

$$\text{Ratio } \frac{\Delta P}{\Delta P'} = \frac{R^n}{\left(\frac{R}{2}\right)^n} = 2^n = 16$$

(e) Cooking oil has a larger viscosity than water. Hence, the value of k will be smaller. For the same value of n , the graph of $\lg Q$ against $\lg r$ in Fig. 6.3 will shift downwards. Similarly, the graph in Fig. 6.2 for cooking oil should scale down along the Q -axis.